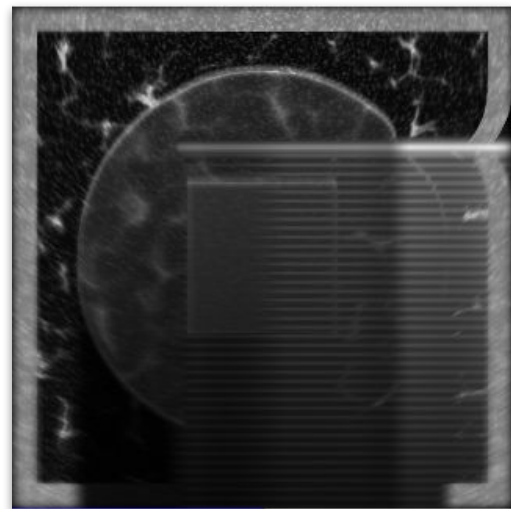
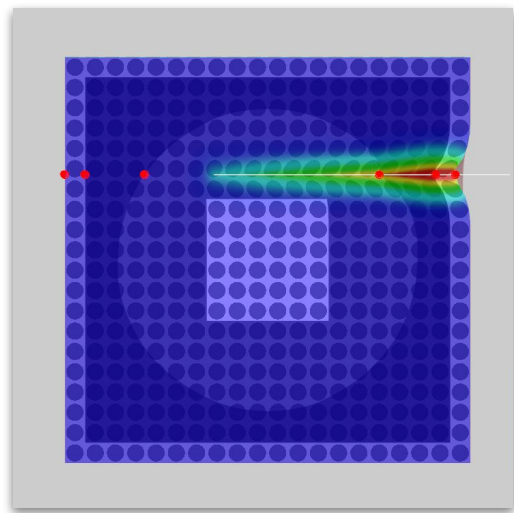


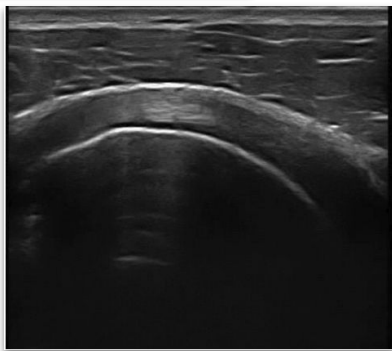
Simulation d'insertion d'aiguille écho-guidée



Conférence JFIG 2022

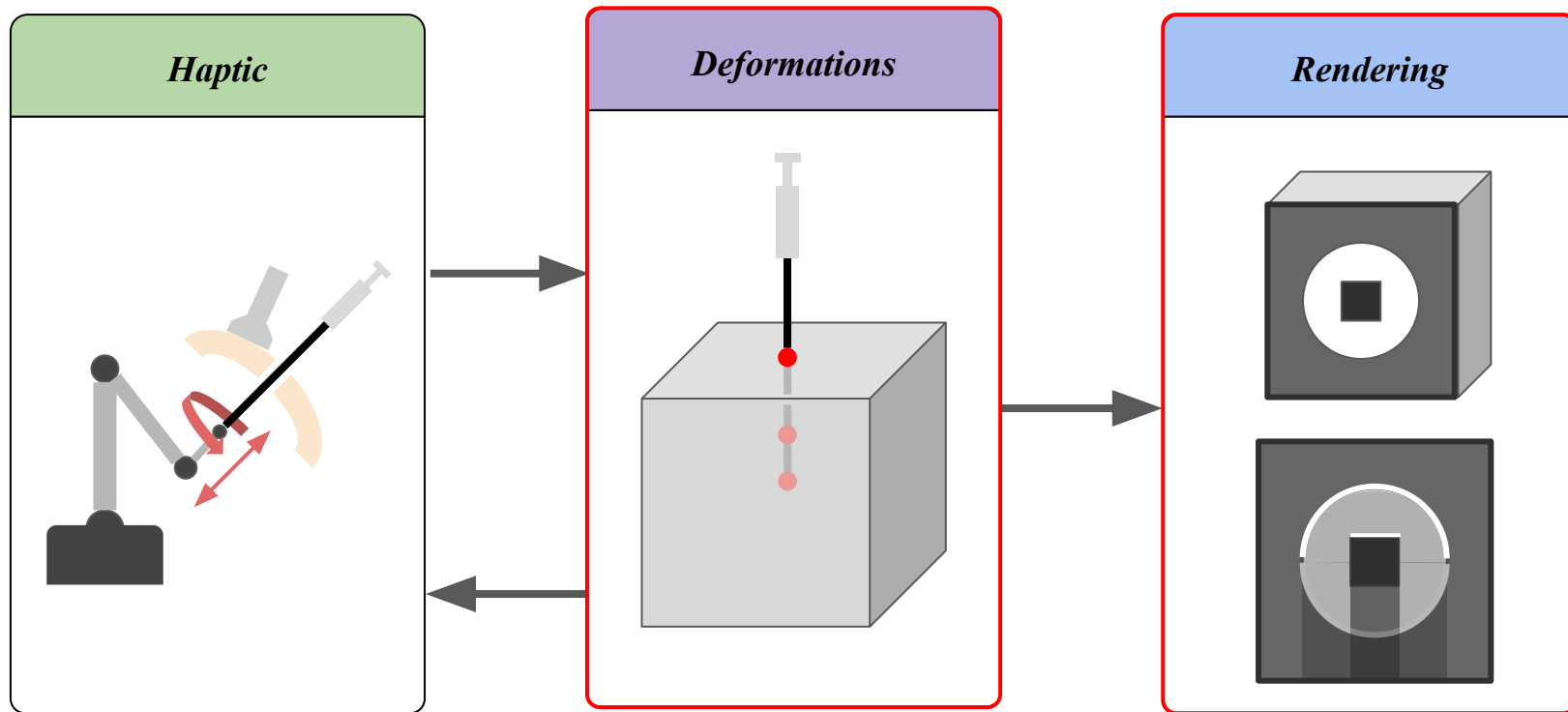
Bastien Saillant, Fabrice Jaillet, Florence Zara

Medical simulator for Rheumatology

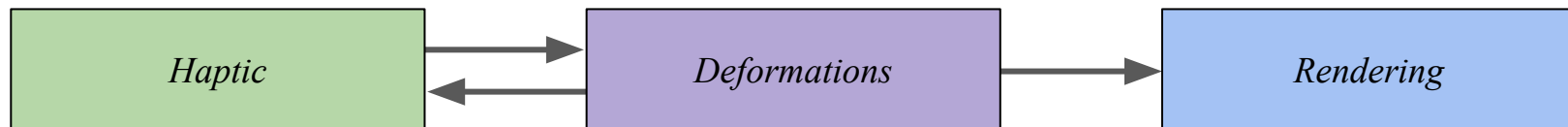


2020 - Charles Barnouin et al: Outil pédagogique de ponction des grosses articulations sous échographie

Pipeline



Contributions



Constraints

Stability

Performance

Realism

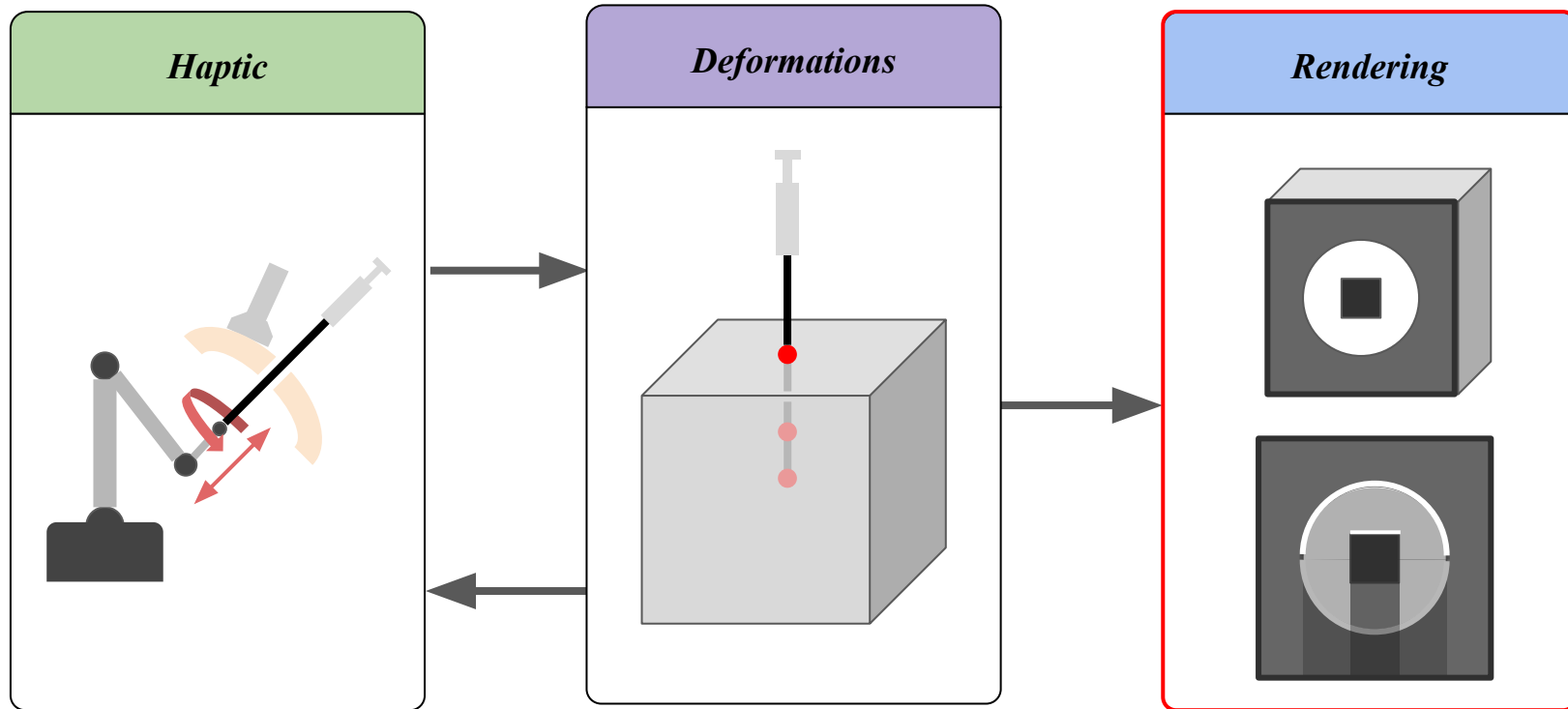


Depth peeling
echography rendering

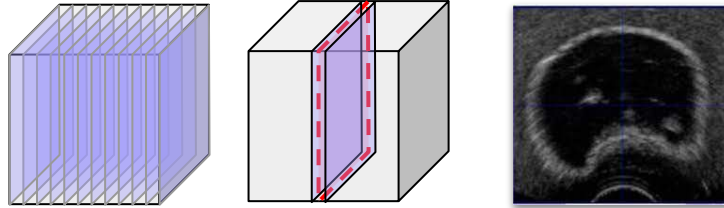
Implicit 3D Materials

XPBD meshless

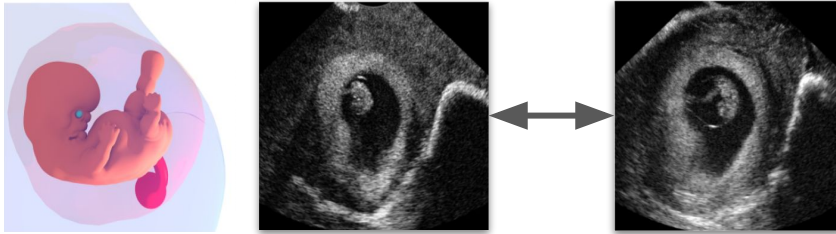
Rendering



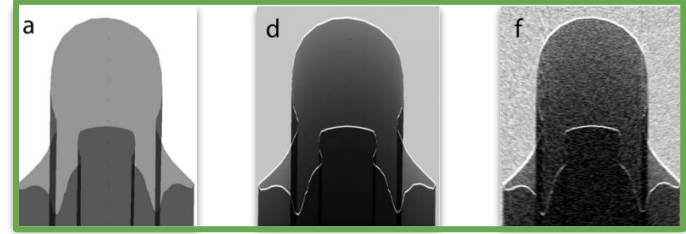
Ultrasound rendering



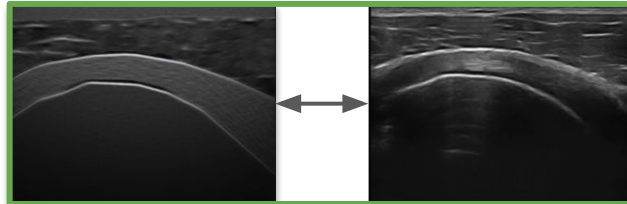
2009 - Sonia-Yuki Selmi: 3D Interactive Ultrasound Image Deformation for Realistic Prostate Biopsy Simulation



2016 - Oliver Mattausch and al: Monte-carlo ray-tracing for realistic interactive ultrasound simulation



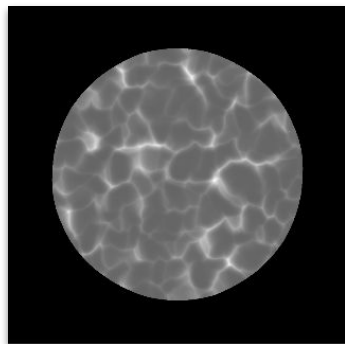
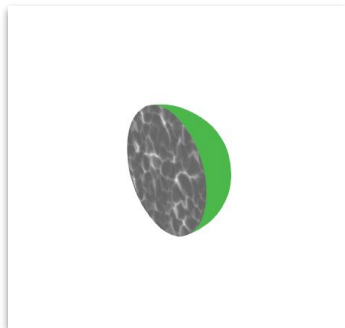
2012 - Yuen C. Law and al: Ultrasound Image Simulation with GPU-based Ray Tracing



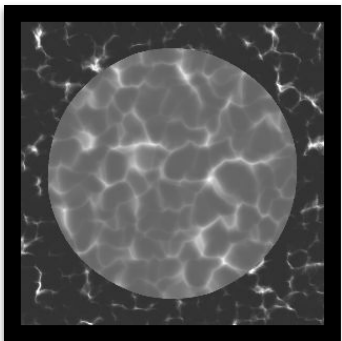
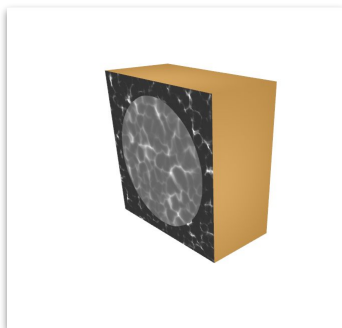
2020 - Charles Barnouin et al: Outil pédagogique de ponction des grosses articulations sous échographie

Echography rendering

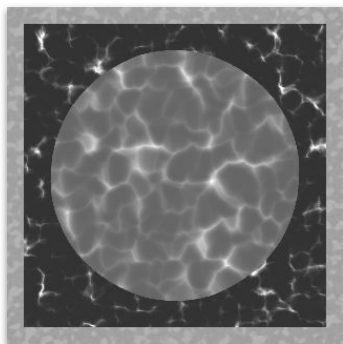
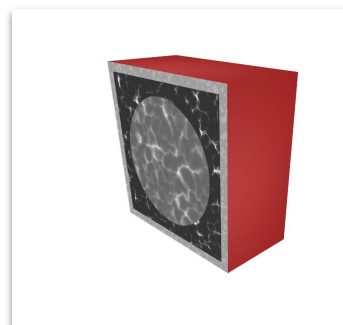
Multi-material cut with *Capping plane solids* method



1 Object



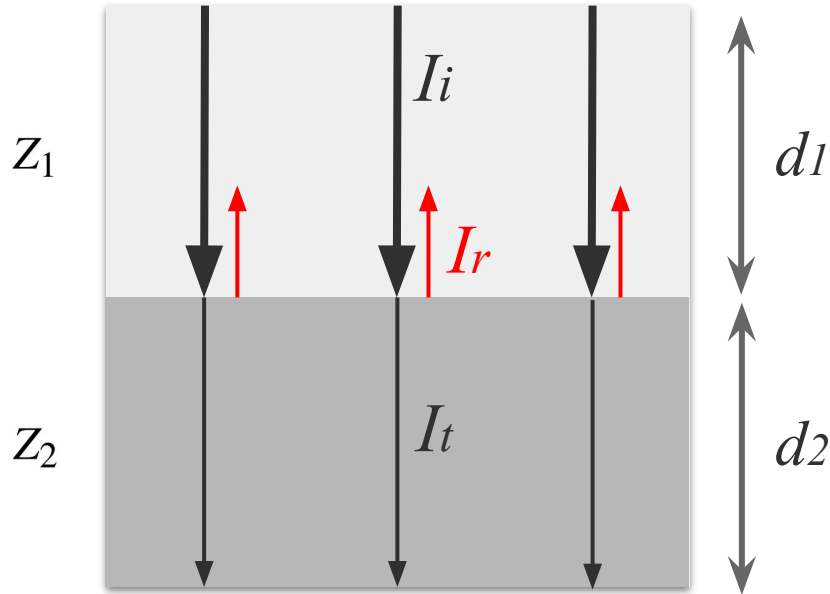
2 Objects



3 Objects

Echographic phenomenon

$$I_{pixel} = I_a \cdot I_t + I_r$$



$$I_r = \left[\frac{Z_1 - Z_2}{Z_1 + Z_2} \right]^2$$

$$I_t = 1 - I_r$$

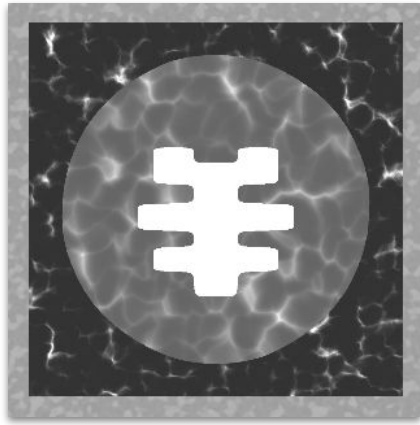
$$I_a = 10^{-\alpha d f / 20}$$

$$\alpha = 20 \cdot \log_{10} e \cdot \beta$$

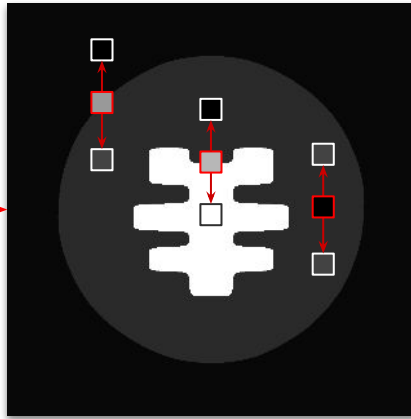
Reflection

$$I_r = I_i \left[\frac{Z_1 - Z_2}{Z_1 + Z_2} \right]^2$$

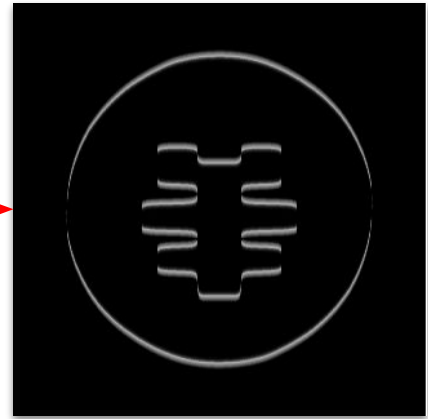
$$I_r = \frac{1}{n} \sum_{j=1}^n \left[\frac{T(x, y - j) - T(x, y + j)}{T(x, y - j) + T(x, y + j)} \right]^2$$



Materials

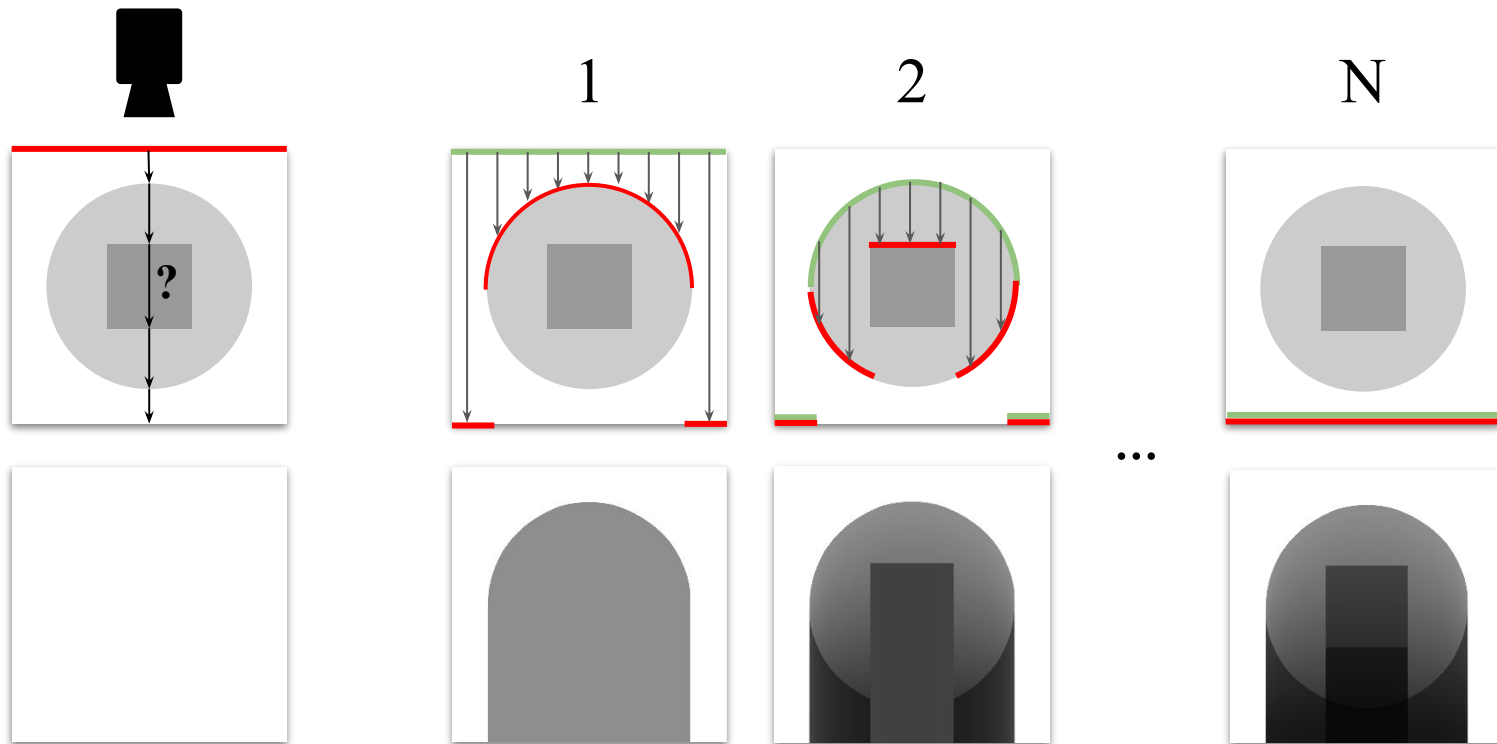


Impedance map



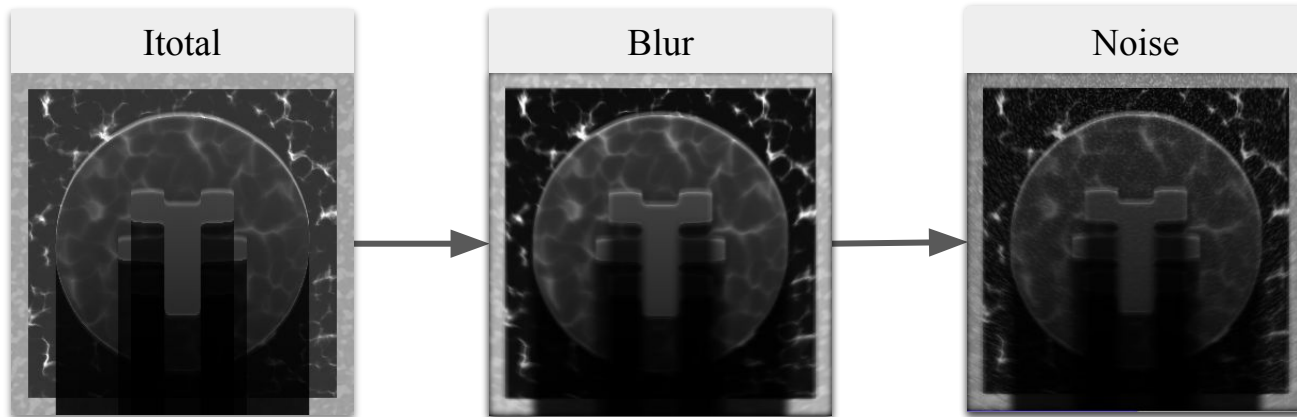
Reflection map

Depth peeling echography rendering

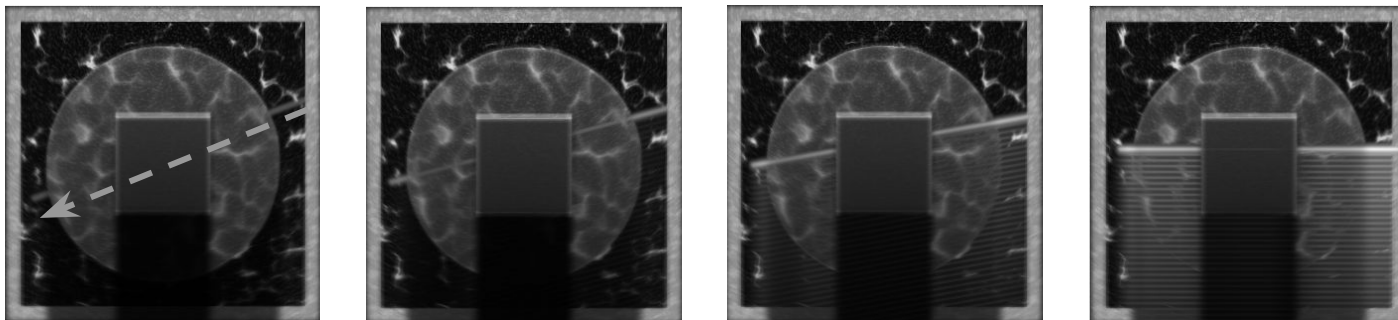


$$I_a \cdot I_t$$

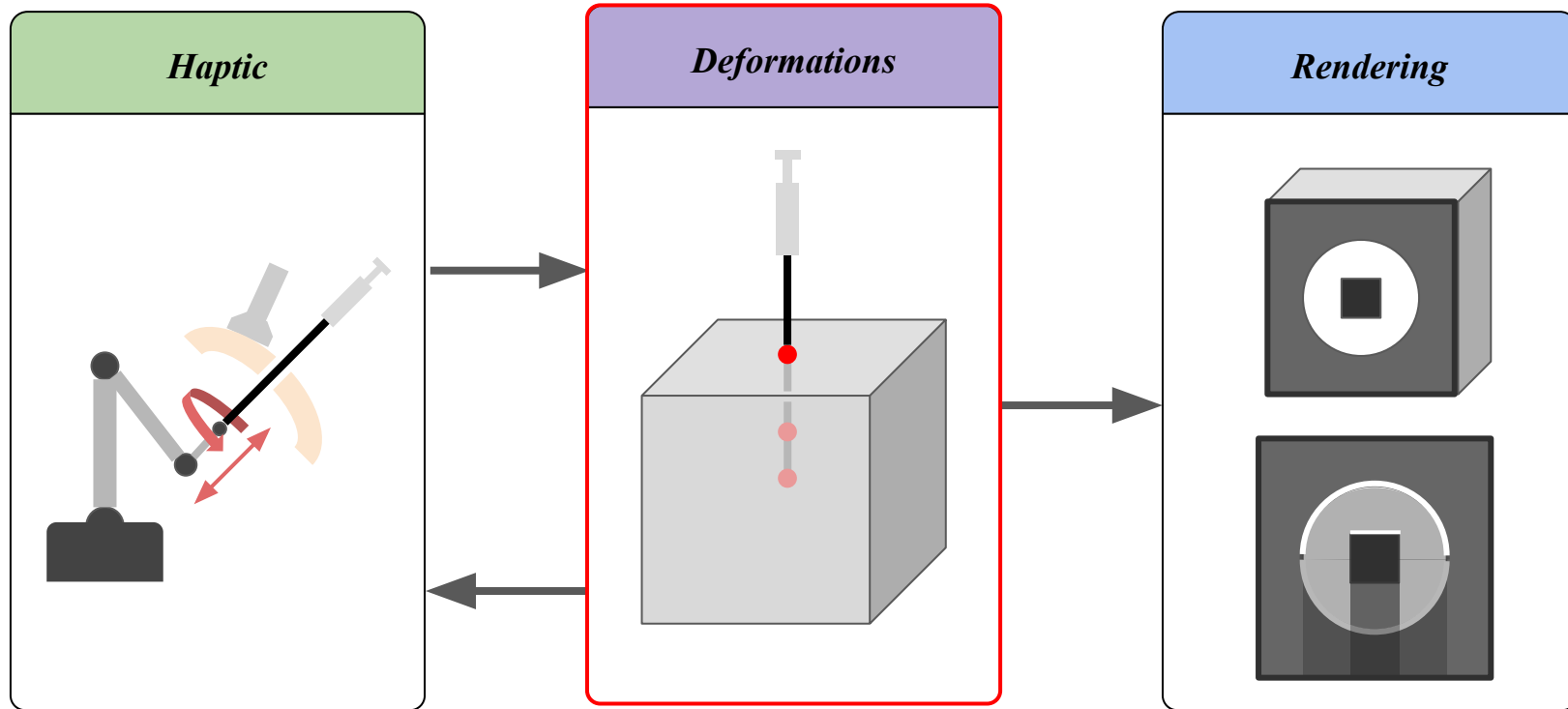
Post processing



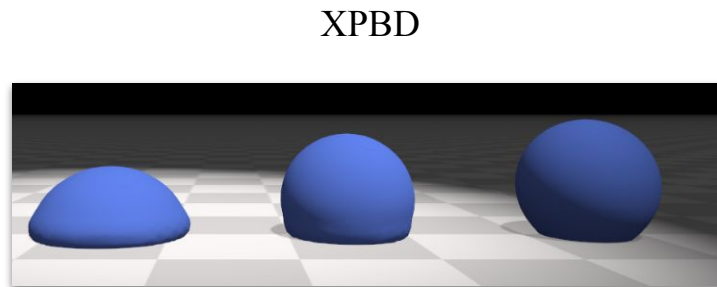
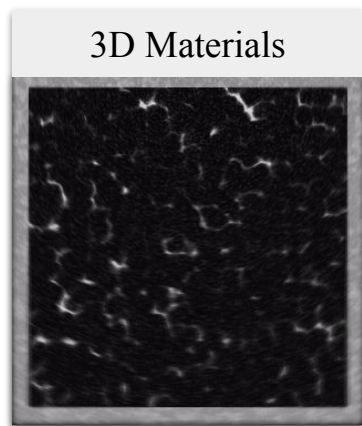
Needle display + Comet Tail



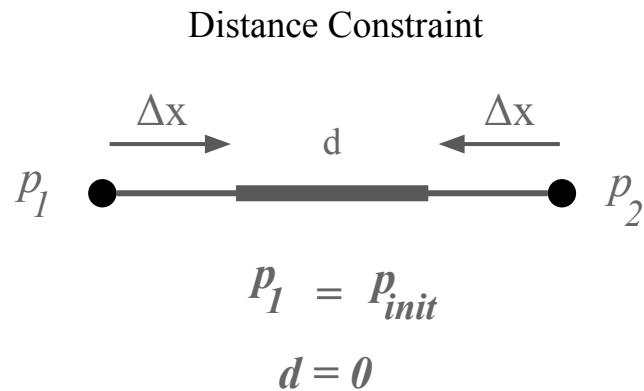
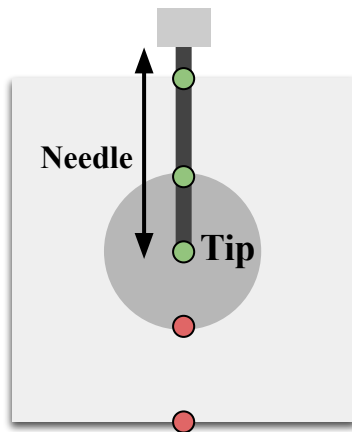
Deformation



Needle simulation



2019 - Miles Macklin et al: XPBD: Position-Based Simulation of Compliant Constrained Dynamics



Needle modelisation

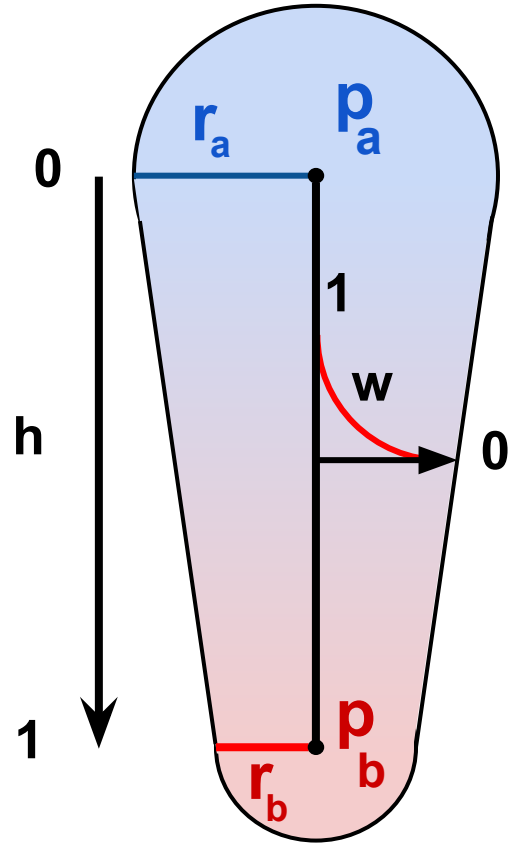
$$h(x) = \frac{\overrightarrow{xp_a} \cdot \overrightarrow{p_b p_a}}{\|\overrightarrow{p_b p_a}\|}$$

$$linear(\vec{u}_a, \vec{u}_b, t) = \vec{u}_a(1 - t) + \vec{u}_b t$$

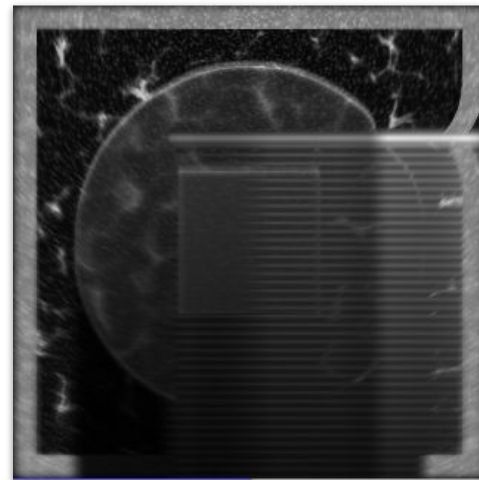
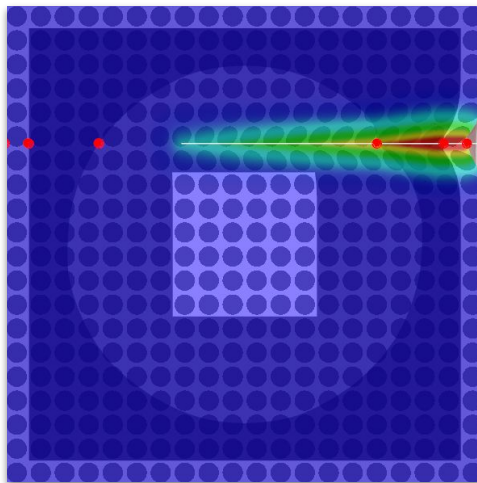
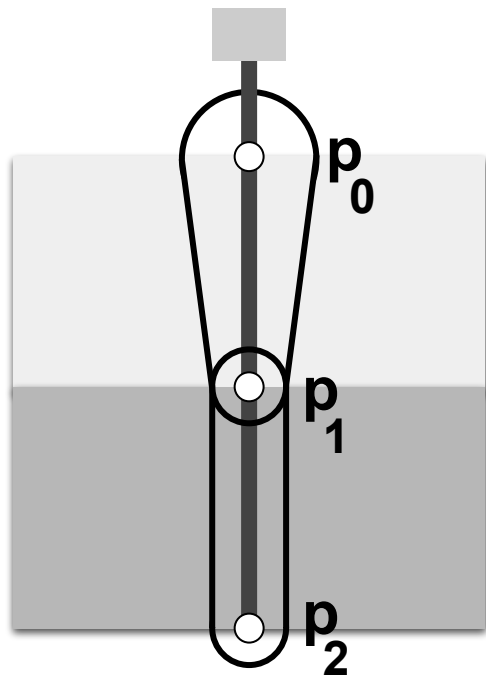
$$d_{cap}(x) = \frac{h(x)(\overrightarrow{xp_a} - \overrightarrow{p_b p_a})}{r_b h + r_a(1 - h(x))}$$

$$w(t) = -t^3 + tx^2 + -tx + 1$$

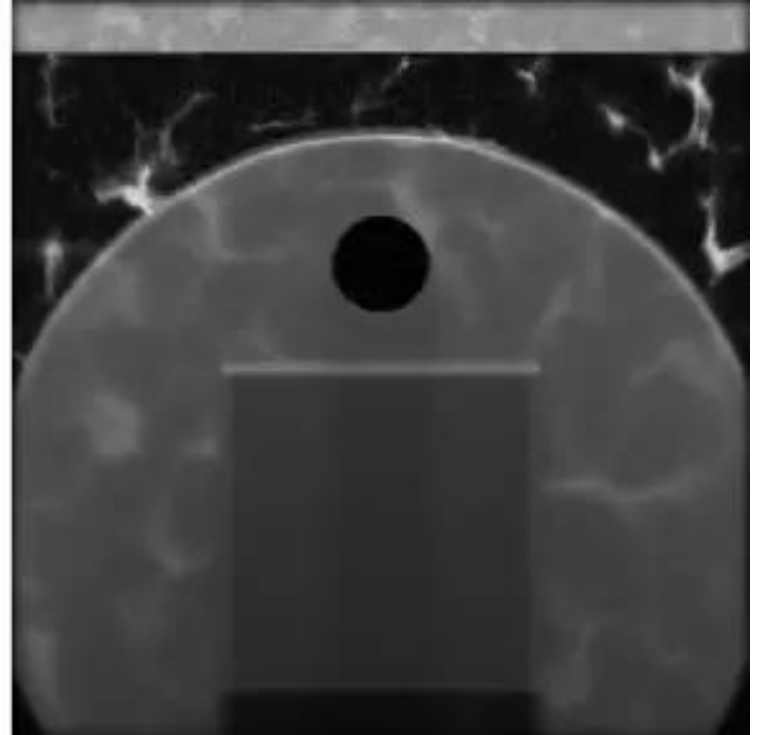
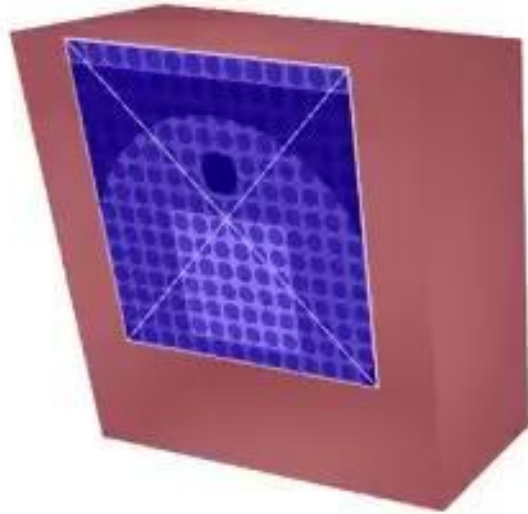
$$g(x) = linear(\vec{u}_a, \vec{u}_b, h(x))w \circ d_{cap}(x)$$



Needle deformation + Rendering



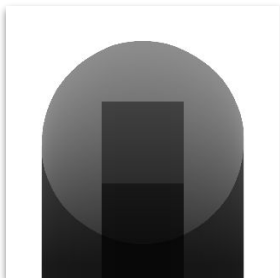
Demo



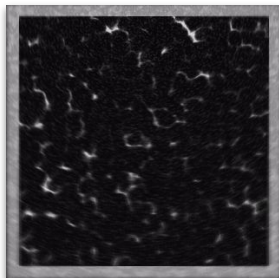
Conclusion



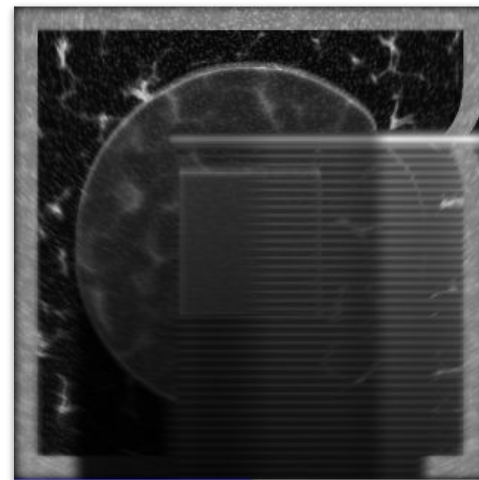
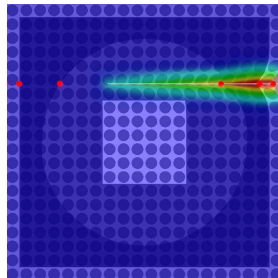
*Depth peeling
echography rendering*



Implicit 3D Materials

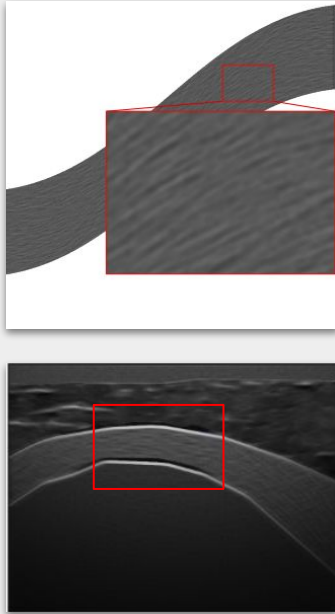


XPBD meshless



Future work

Fiber orientation



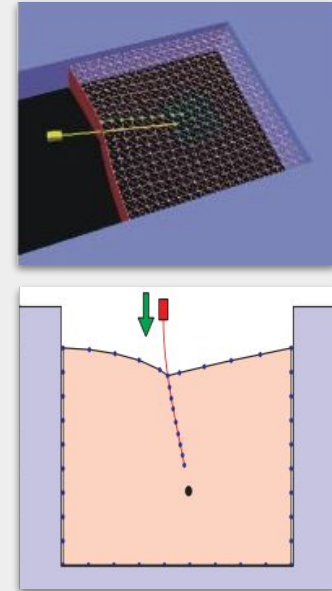
2020 - Charles Barnouin et al: Outil pédagogique de ponction des grosses articulations sous échographie

Raytracing ++



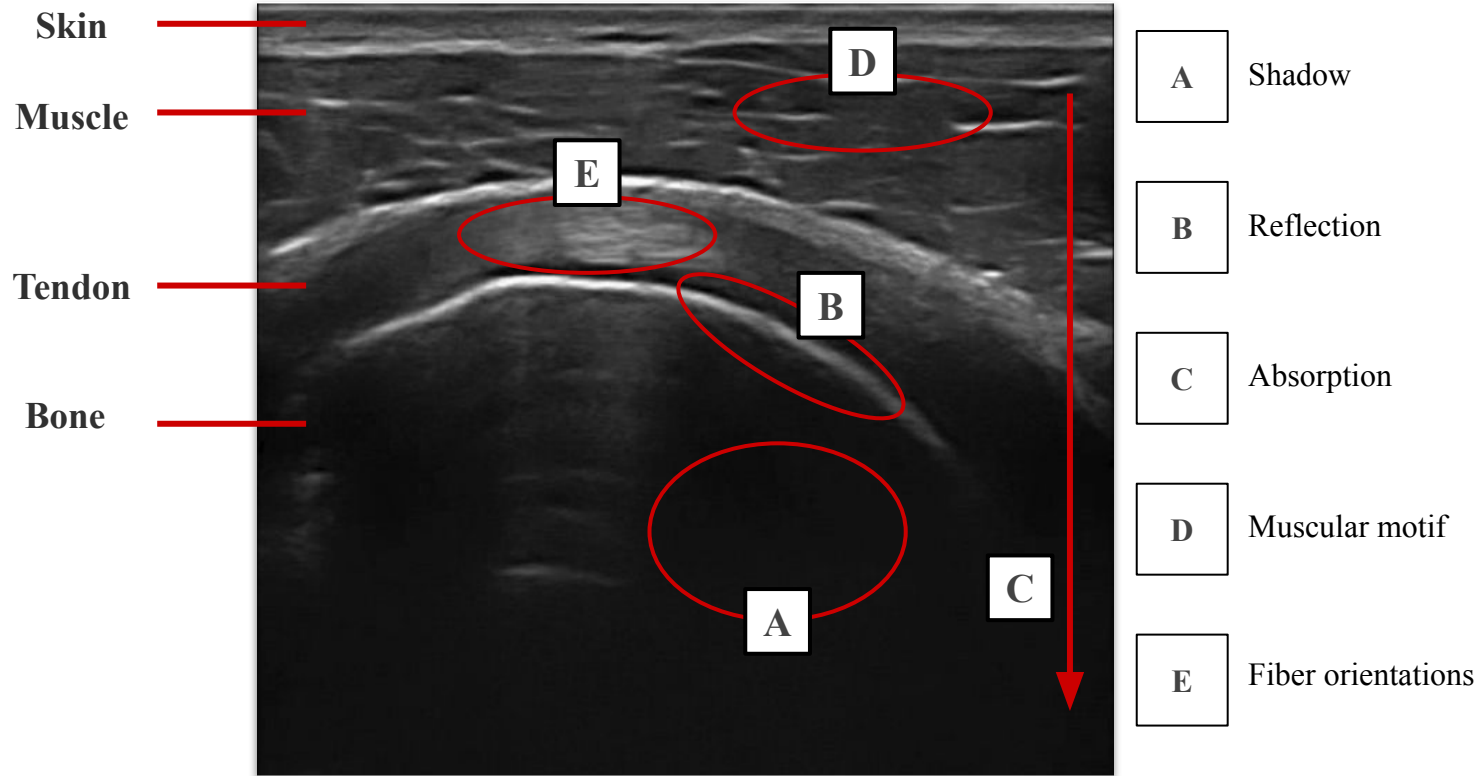
2016 - Oliver Mattausch et al: Monte-carlo ray-tracing for realistic interactive ultrasound simulation

FEM Method



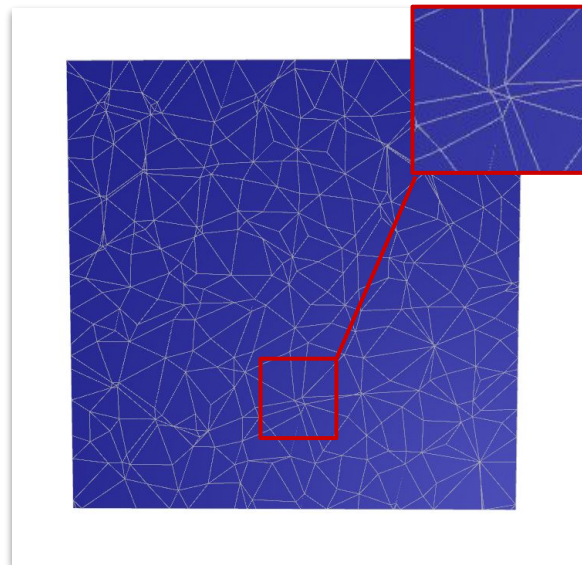
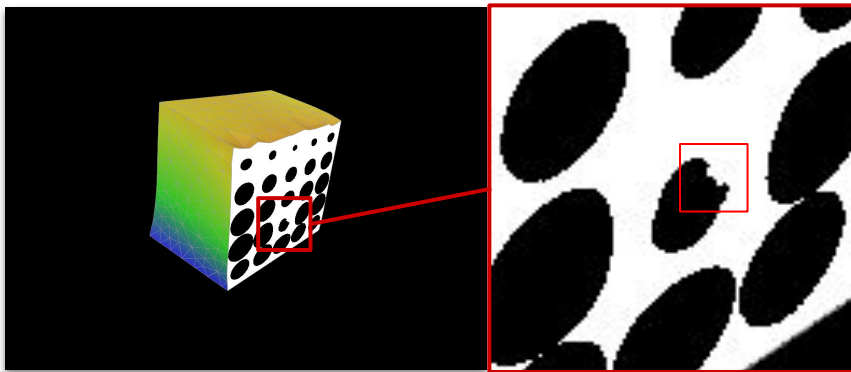
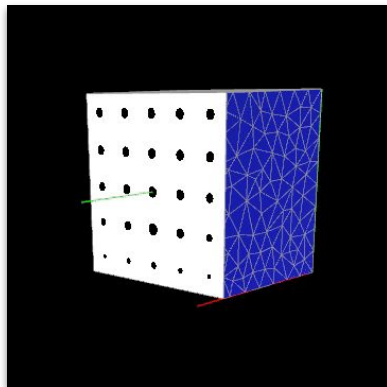
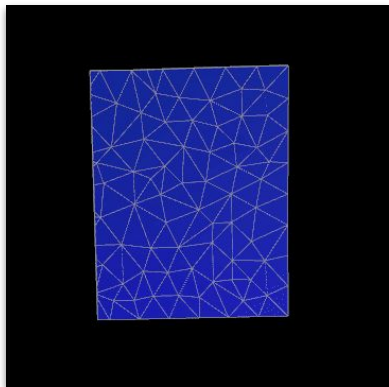
2002 - S.P. DiMaio et al: Needle Insertion Modelling for the Interactive Simulation of Percutaneous Procedures

Phénomènes échographiques

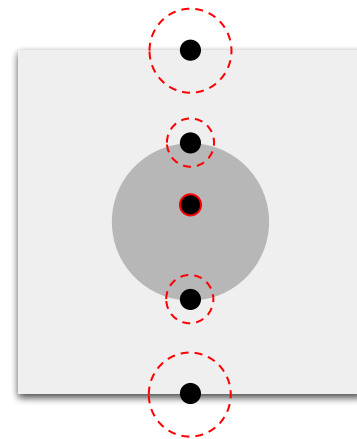
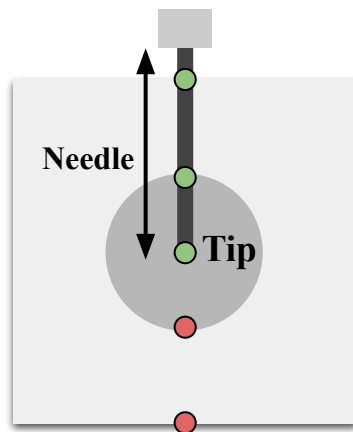
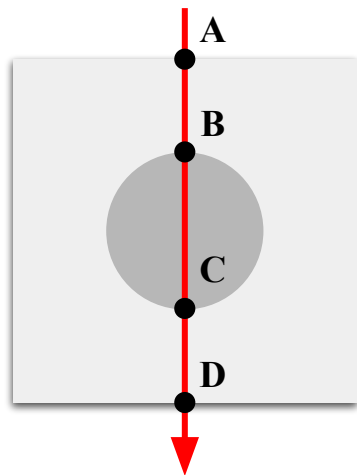


Ecography of a shoulder

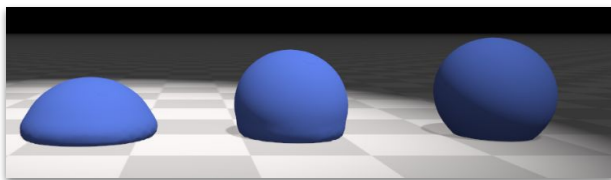
FEM



Needle simulation



XPBD



$$p_1 = p_{init}$$

$$d = 0$$